

HDD-ESA: Engineering Systems Architecture

A Practical Guide to History-Dependent Causal Testing and Testability-by-Design

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August 2026

In short: many control-system problems that look like tuning drift, intermittent faults, or “just how the equipment behaves” are actually history-dependent effects — the system’s past trajectory, not just its current state, is shaping its behavior. This guide gives five concrete tests to determine whether that is happening in a given system, isolate what is causing it, and design new systems so the answer is cheap to obtain. Missing this distinction means retuning controllers that were never the problem, replacing hardware that was never faulty, or chasing intermittent faults with no fixable root cause.

Companion document to *HDD-ISA — AI Architectures for Causal Discriminations* (Taotuner, 2026b) and the *History-Dependent Dynamics (HDD)* framework (Taotuner, 2026a). HDD-ISA applies the same methodology to AI architectures; HDD-ESA applies it to physical control systems — motion, thermal, fluid, chemical, and power — with five causal-discrimination tests, design rules for building testability into new systems, and per-substrate implementation guides. The two documents share reporting categories (§9).

Methodological status. This is a diagnostic and design-for-testability method, not a general theory of control-system architecture. It determines whether history dependence, path dependence, feedback mediation, internal-state dependency, or model-based causal control are present in a system, and how to build new systems so that determination is cheap to make. It does not replace standard control-engineering practice (Åström & Murray, 2008) for choosing a control topology. Numeric thresholds throughout §§3–7 (10%, 15%, 20%, and the noise-floor multiplier in T2) are operational defaults, not universal constants — tune them to the noise characteristics and stakes of your own system.

On the worked examples. The worked examples in §§3–7 are illustrative: they show how each test’s interpretation table applies to plausible numeric outcomes, not measurements from real systems. No hardware, lab, or dataset underlies these numbers. Report your own instrumentation, environmental conditions, sample sizes, and raw-data availability when you apply this guide in practice.

1. Scope

Target systems: motion control (motors, servos, robots); thermal processes (ovens, heat exchangers, HVAC); fluid systems (pumps, valves, hydraulic actuators); chemical

processes (reactors, mixers, distillation columns); power systems (converters, drives, grids).

Requirements: a plant; sensors (position, temperature, pressure, flow, current, voltage); actuators; a controller (PID, MPC, or any regulator); a data logger.

Out of scope: AI systems and machine learning models (see HDD-ISA); philosophical questions about consciousness; software-only simulation.

2. The Five Tests

Test	Question	Requires
T1	Does history improve prediction?	Sensor logs, system identification
T2	Does past trajectory cause future behavior?	Two trajectories to the same setpoint, confound control
T3	Is the trajectory effect mediated by a specific feedback pathway?	Ability to open that pathway, plus a sham comparison pathway
T4	Does the system depend on an internal state estimate, and is that dependency self-specific?	Internal state estimate, plus a matched decoy channel
T5	Does the system use a self-inclusive model causally?	A model-based controller (MPC, LQR)

Table 1. The five HDD-ESA tests.

Each test targets a logically distinct construct: passing T1 does not imply T2, passing T2 does not imply T3, and so on. §12.2's suggested test order is a resource-allocation heuristic, not a claim of logical dependency — a system can in principle pass a later test while failing an earlier one.

T4 and T5 use narrow, functional definitions of “self-reference” and “self-model,” defined precisely in §6 and §7. Passing them establishes a specific, testable engineering property, not a claim about self-awareness or agency.

3. T1 — History-Dependent Prediction

A positive result establishes that including history improves prediction beyond the current measured state — not that the system “lacks a Markovian description.” Any system has a Markovian description in a suitably enlarged state space (Åström & Murray, 2008); T1 detects that $y(t)$ and $u(t)$ are an incomplete observation of that larger state (unmodeled thermal lag, actuator elasticity, hysteresis), not memory in any stronger sense.

Protocol. Log $y(t)$ and $u(t)$ at fixed rate T_s . Fit Model 0 (order 1: $y(t+1) = f(y(t), u(t))$) and Model 1 (order n , $n \approx 3 \times \text{dominant time constant} / T_s$), matched in parameter

count, via system identification (Ljung, 1999; Söderström & Stoica, 1989). Validate on a chronological 80/20 held-out split.

Significance. Use the F-test as the primary criterion — $F = [(SSE_0 - SSE_1)/(df_0 - df_1)] / [SSE_1/df_1]$ — and report AIC/BIC ($AIC = n \cdot \ln(SSE/n) + 2k$; $BIC = n \cdot \ln(SSE/n) + k \cdot \ln(n)$) as a complementary check, not an alternative pass condition: AIC and BIC penalize complexity differently and answer a model-selection question, not a hypothesis-testing question, so treating “F-test $p < 0.05$ OR AIC/BIC improves” as an either/or criterion inflates the effective false-positive rate.

F-test	AIC/BIC	Result
$p < 0.05$	Both improve by > 2	Supported
$p < 0.05$	Disagree or don't improve	Uninterpretable — investigate before reporting a pass
$p \geq 0.05$	—	Negative Evidence

Table 2. T1 interpretation.

Illustrative example (DC motor, 10,000 samples at 1 kHz): order-3 vs. order-1 model of speed vs. voltage; MSE improves 29.8%, F-test $p < 0.001$, $\Delta AIC = 15.2$, $\Delta BIC = 11.0$ — **Supported**, consistent with unmodeled electrical and mechanical time constants rather than evidence of anything beyond an incomplete first-order observation.

4. T2 — Causal Trajectory Dependence

A bare comparison of two trajectories reaching the same setpoint does not establish causal trajectory dependence: ambient drift, incomplete return to a common initial state, and accumulated thermal or magnetic energy can each produce an apparent trajectory effect unrelated to the trajectory itself. The protocol therefore requires randomized/interleaved trial order; a verified, tolerance-banded return to baseline before each trial; logged environmental covariates (ambient temperature, supply voltage) included via ANCOVA rather than a plain t-test; and confirmation that accumulated energy state is matched at trial start.

Protocol. Design two trajectories (e.g., fast vs. slow ramp) to the same setpoint r ; apply a standard perturbation after each; record the response (overshoot, settling time) over ≥ 5 –10 repetitions per trajectory in randomized order; analyze by ANCOVA with logged covariates.

Significance. Set the practical-effect threshold relative to the system's own noise floor (σ_{noise} , from repeated same-trajectory trials) rather than a fixed percentage, since a universal threshold is not meaningful across systems of different scale and tolerance:

Result	Interpretation
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Result	Interpretation
ANCOVA $p < 0.05$ AND effect $> k \cdot \sigma_{\text{noise}}$ ($k = 3$ as a default)	Supported
$p \geq 0.05$, or effect below threshold	Negative Evidence
Confound controls could not be fully implemented	Non-Identifiable — report which controls were missing

Table 3. T2 interpretation.

Illustrative example (oven, setpoint 200 °C): fast vs. slow ramp, ambient temperature logged as covariate, 10 randomized repetitions each, 30-minute baseline soak between trials; overshoot difference 7.2 °C against $\sigma_{\text{noise}} = 0.9$ °C ($\approx 8\times$), ANCOVA $p < 0.001$ — **Supported**.

5. T3 — Feedback Mediation

Opening a feedback loop changes system dynamics — bandwidth, damping, noise sensitivity — regardless of whether that loop specifically mediated the T2 effect. A reduction in divergence after opening the candidate loop is therefore necessary but not sufficient evidence of loop-specific mediation; almost any sufficiently large dynamical perturbation can produce a similar reduction. The protocol requires a sham condition: a different loop, or a partial change to the same loop, perturbed by a matched dynamical magnitude, without being the hypothesized mediator.

Protocol. Measure the baseline T2 effect (loop intact, $n \geq 5$); open the candidate loop (zero gain, precomputed open-loop profile, or physical bypass) and re-measure ($n \geq 5$); run the matched sham condition and measure the same effect ($n \geq 5$).

Result	Interpretation
Candidate reduces the effect ($p < 0.05$) by significantly more than the sham	Supported — loop-specific mediation
Candidate and sham produce comparable reductions	Uninterpretable as loop-specific — consistent with a generic dynamical artifact
No reduction under either condition	Negative Evidence

Table 4. T3 interpretation.

Illustrative example (PID oven): baseline divergence 7.2 °C; PID opened $\rightarrow$ 2.1 °C (71% reduction, $p < 0.01$); sham (rate-limited actuator, matched bandwidth change) $\rightarrow$ 6.4 °C (11% reduction, n.s.) — **Supported**.

6. T4 — Internal-State Dependency and Self-Referential Specificity

Showing that perturbing an internal state estimate changes control output more than perturbing a matched external variable establishes only that the controller depends on the state estimate it was designed to use — true, by construction, of essentially any competently designed closed-loop controller. The “self” vs. “non-self” label is stipulated by the engineer, not discovered by the test. This guide therefore splits T4 into a weak form (dependency) and a strong form (specificity), following the decoy-comparison logic used for the analogous construct in HDD (Taotuner, 2026a, §20.4) and HDD-ISA.

T4-Weak. Inject a bias into the internal state estimate $\hat{x}$ (defined by $d(\hat{x})/dt = A \cdot \hat{x} + B \cdot u + L \cdot (y - C \cdot \hat{x})$) and confirm the control output responds ($n \geq 5$, $p < 0.05$). A pass establishes internal-state dependency only, and should not be described as self-reference without that qualifier.

T4-Strong. Construct a matched decoy channel: a variable of similar magnitude, update rate, and information content to $\hat{x}$, but not derived from or fed back into the system’s own regulation — an independent signal injected through an auxiliary, non-regulatory input. Inject matched-magnitude perturbations into the genuine estimate and the decoy, in randomized order ($n \geq 5$ each), and compare effect sizes.

Result	Interpretation
Genuine effect substantially exceeds decoy effect (report ratio and CI)	Supported — dependency specific to self-referential content
Genuine and decoy effects indistinguishable	Negative Evidence — explained by generic information-bearing input, not self-reference

Table 5. T4-Strong interpretation.

A system passing T4-Weak but failing T4-Strong should be reported as “internal-state dependent, no evidence of self-referential specificity” — not folded into an ambiguous single “T4 passed.”

Illustrative example (BLDC motor, FOC control): genuine estimate = rotor angle $\hat{\theta}$ from a Luenberger observer (Luenberger, 1964); decoy = matched independent signal via an unused input. T4-Weak: $+5^\circ$ bias $\rightarrow$ torque ripple $+15.2 \pm 2.1\%$ ($p < 0.001$), **Supported**. T4-Strong: matched decoy perturbation $\rightarrow +2.1 \pm 1.8\%$ (n.s.); ratio 7.2 (95% CI 4.1–11.8) — **Supported**.

7. T5 — Model-Based Predictive Control

“Causal self-model” here means, specifically, a model that includes a representation of the system’s own dynamics and is used causally, not descriptively, to generate control action — standard MPC terminology (Camacho & Bordons, 2007) under a name

consistent with the HDD construct hierarchy. Passing T5 is an engineering statement about a control architecture, not a claim about a new category of self-modeling.

Three components must all pass:

Component	Test	Pass condition
A: Self-prediction	Model predicts future states	Held-out error < 10% of signal range
B: Counterfactual prediction	Model predicts effect of a different input	Held-out error < 15%
C: Causal policy influence	Model modification changes control output	Output changes > 20%, $p < 0.05$

Table 6. T5 three-component criterion.

The 10%/15% thresholds follow standard model-validation practice (Ljung, 1999); the 20% threshold for Component C is the author's operational choice (roughly twice typical control-output variance) and should be adjusted to the noise floor of the system under test.

Illustrative example (chemical reactor with MPC): Component A: 6.2% error, pass. Component B: 11.8% error, pass. Component C: +20% rate constant → output changes $35.4 \pm 4.2\%$, $p < 0.001$, pass — **Supported**.

8. Design Rules for Testability

Each test is cheaper and safer when the system is architected for it from the outset, rather than retrofitted:

Test	Design rule
T1	Log raw, unfiltered signals at $\geq 10\times$ the inverse of the shortest time constant of interest, synchronized timestamps, no in-path smoothing that could mask or manufacture apparent history dependence; buffer depth $\geq 3\times$ dominant time constant.
T2	Programmatic trajectory-select mode; automated randomized-order sequencer; automated baseline-return check; ambient-condition sensors provisioned as standard instrumentation.
T3	Accessible bypass/manual-override per candidate loop (software or physical switch); at least one auxiliary parameter suitable as a matched sham condition.
T4	Internal estimator output exposed as a separately loggable and injectable signal; a decoy input channel with no path into regulation, provisioned at design time — the single highest-value item in this table, since building it after the fact is usually the most expensive part of running T4-Strong.
T5	Supervisory interface for controlled, logged, reversible model-parameter

Test	Design rule
	injection, rather than direct edits to controller code during testing.

Table 7. Design rules for testability.

9. Reporting Categories

Shared with HDD-ISA for consistency across the series:

Category	Meaning
Supported	The evidentiary bar was met.
Negative Evidence	Tested under adequate conditions; bar not met.
Uninterpretable	Internally inconsistent results (e.g., F-test and AIC/BIC disagree; candidate and sham indistinguishable) requiring investigation before a verdict.
Non-Identifiable	Required controls or instrumentation (decoy channel, covariate logging) could not be implemented.

Table 8. Reporting categories, consistent with HDD-ISA.

10. Interpretation

Profile	Interpretation	Design implication
T1 Supported, T2 Negative Evidence	Memory exists, trajectory doesn't matter	Higher-order model; standard setpoint control
T1, T2 Supported; T3 Negative Evidence	Trajectory matters, not through the tested pathway	Investigate feedforward, hysteresis, or an untested pathway
T1-T3 all Supported	Trajectory matters through the identified feedback pathway	PID with anti-windup; monitor integral saturation
T4-Weak Supported, T4-Strong Negative Evidence	Ordinary feedback dependency, no self-referential specificity	Report as such; avoid self-reference language
T4-Strong Supported	Dependency specific to self-referential content	Observer is a critical, non-interchangeable component; prioritize fault detection
T5 Supported	Self-inclusive model used causally (§7 sense)	MPC/adaptive control is engineering-justified

Table 9. Consistent result profiles and their implications.

Common failure modes. T1: overfitting or large-n significance without practical gain (false positive); low excitation (false negative). T2: unlogged confounds (false positive); insufficient perturbation (false negative). T3: generic dynamical perturbation mistaken for loop-specific mediation, mitigated by the sham control (false positive); redundant feedback loops (false negative). T4: poorly matched decoy (false positive); insufficient sensitivity (false negative). T5: correlation mistaken for causation, mitigated by Component C (false positive); insufficient excitation (false negative).

11. Substrate Notes

Substrate	Instrumentation	Candidate T3 loop / T2 trajectory	T4/T5 notes
Motion control (motors, servos, robots)	Encoder/tachometer, current sensing	Inner current loop vs. outer velocity/position loop; trapezoidal vs. S-curve profiles (usually already in firmware)	FOC rotor-angle/flux observers are a natural T4 channel; MPC/LQR increasingly common for T5
Thermal (ovens, HVAC, heat exchangers)	Thermocouples/RTDs, mandatory ambient sensor	PID vs. rate-limited actuator (sham); startup ramp rate	Baseline-return times can be long (minutes–hours) — budget accordingly; T4/T5 less common outside advanced HVAC/industrial ovens
Fluid (pumps, valves, hydraulics)	Flow/pressure sensors, valve position	Pressure feedback loop; fast vs. slow valve actuation	Hysteresis from seals/stiction can mimic T1's observational gap — distinguish via explicit hysteresis modeling (Mayergoyz, 1991); validate T3 in simulation

Substrate	Instrumentation	Candidate T3 loop / T2 trajectory	T4/T5 notes
Chemical (reactors, columns)	Temperature/pressure/composition, often limited observability	Batch startup profile; long baseline-return between batches	first given cavitation risk Safety: T3/T4 interventions carry runaway/pressure- re-excursion risk — validate in simulation and follow site process-safety and change- control procedures before live testing. T5 is often cheap here since MPC and parameter- injection interfaces frequently already exist.
Power (converters, drives, grids)	Voltage/current/frequency, protection-relay status	Protective loops (frequency/voltage)	Safety: do not open protection loops on live infrastructure; use a test bench, digital twin, or islanded microgrid. Fast switching dynamics require care with T1's sampling-rate rule (§8) to avoid aliasing. Grid- forming inverters with state estimators suit T4; MPC- based converter

Substrate	Instrumentation	Candidate T3 loop / T2 trajectory	T4/T5 notes control suits T5.
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Table 10. Substrate-specific implementation notes.

12. Implementation Tools and Checklist

12.1 Tools

MATLAB: System Identification Toolbox (ARX/ARMAX, F-test, AIC/BIC); Control System Toolbox (PID, LQR, observers, Kalman filters). Python: `control`, `sippy` (ARX/ARMAX), `scipy` (filtering, statistical tests). Open source: github.com/python-control/python-control; github.com/CPCLAB-UNIPY/SIPPY; scipy.org.

12.2 Checklist

Prerequisites: signal logging per §8's sampling rule; trajectory-select with randomization and baseline-return check; loop bypass plus sham pathway; state-estimate injection plus decoy channel; model-parameter injection interface.

Suggested order (resource-allocation heuristic, not logical dependency): T1 → T2 (with confound controls) → T3 (with sham) → T4-Weak → T4-Strong (if resources allow) → T5.

Reporting: use the four categories in §9 for every test, noting which controls, sham conditions, or decoy channels were or were not implemented.

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